

A Proposed Construction of a Complex Structure on the Six-Sphere

A streamlined detailed reconstruction of the S^6 argument

Reconstructed from the source manuscript [1]

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Abstract

We reconstruct only the chain of arguments in the source manuscript [1] that is claimed to produce an integrable complex structure on the standard smooth six-sphere. The retained material consists of the period family over the $(3, 4, \infty)$ orbifold, its three analytic fillings, the fundamental-group calculation, the integral nearby-cycle and Leray calculations, and recognition of the resulting manifold. All general parameter families, duplicated matrix appendices, and invariants unused by this conclusion are removed. The delicate cusp-properness and integral-saturation arguments remain explicit.

Status of the argument

This is a proposed solution reconstructed from [1], not an independently established resolution of the Hopf problem. The argument below is therefore conditional in the following precise sense: if its analytic quotient, nearby-cycle, and signed transgression computations are correct, then its final conclusion follows. The proof is written so that those load-bearing computations can be checked directly.

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1 Claim, conventions, and logical dependencies

Theorem 1.1 (Target claim). *The standard smooth six-sphere S^6 admits an integrable complex structure.*

The proof has four logical blocks, divided below into an analytic construction and an integral-topological verification.

1. Construct a compact complex threefold $f : X \rightarrow B$ over

$$B = \mathbb{P}^1, \quad \Sigma = \{p_0, p_1, p_2\}, \quad (m_1, m_2) = (3, 4)$$

whose general fiber is a complex two-torus, whose fiber over p_0 is reduced with normal crossings, and whose fibers over p_1, p_2 have multiplicities three and four, respectively.

2. Choose fixed logarithmic translations at the two elliptic fibers and use van Kampen's theorem to prove $\pi_1(X) = 1$.
3. Use the integral Leray spectral sequence and nearby cycles at the cusp to prove

$$H_k(X; \mathbb{Z}) \cong H_k(S^6; \mathbb{Z}).$$

4. Hurewicz–Whitehead gives $X \simeq S^6$, and $\Theta_6 = 0$ gives $X \cong_{\text{diff}} S^6$. Transporting the holomorphic atlas of X along this diffeomorphism produces an integrable complex structure on S^6 .

Paths and base loops are composed from left to right, and all three small meridians are oriented clockwise. Reversing all three orientations changes every boundary coefficient by the same sign, so the final conclusion is unchanged.

2 Monodromy and the cusp lattice

2.1 The four-dimensional lattice and the dual action

Let

$$V = \mathbb{Z}^4 = \langle \gamma, u, w, \delta \rangle, \quad \Lambda = V^* = \langle \hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta} \rangle.$$

Define the following matrices, whose columns are the images of the basis vectors:

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$T_1^3 = T_2^4 = I, \quad T_0 := (T_1 T_2)^{-1} = I + N, \quad N^2 = 0,$$

as well as

$$N\gamma = Nu = 0, \quad Nw = -u, \quad N\delta = \gamma, \quad \ker N = \text{Im } N = \langle \gamma, u \rangle.$$

On Λ , set

$$A_j = (T_j^{-1})^t \quad (j = 1, 2), \quad M_0 = (T_0^{-1})^t.$$

Then

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \quad M_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

and $A_1 A_2 M_0 = I$.

Set

$$\Lambda_{\text{tor}} := \langle \widehat{w}, \widehat{\delta} \rangle, \quad \bar{\Lambda} := \Lambda / \Lambda_{\text{tor}} = \langle \bar{\gamma}, \bar{u} \rangle.$$

Since

$$(M_0 - I)(a\widehat{\gamma} + b\widehat{u} + c\widehat{w} + d\widehat{\delta}) = b\widehat{w} - a\widehat{\delta}$$

we have

$$\ker(M_0 - I) = \text{Im}(M_0 - I) = \Lambda_{\text{tor}}.$$

Thus $M_0 - I$ induces the unimodular isomorphism

$$B_0 : \bar{\Lambda} \xrightarrow{\sim} \Lambda_{\text{tor}}, \quad B_0 \bar{\gamma} = -\widehat{\delta}, \quad B_0 \bar{u} = \widehat{w}, \quad [B_0] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

2.2 Invariants, coinvariants, and twisting vectors

Set

$$\varepsilon = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \quad \varepsilon' = \widehat{\gamma} + 3\widehat{u} - 3\widehat{w}.$$

Solving $(A_j - I)\lambda = 0$ gives

$$\Lambda^{A_1} = \langle \varepsilon, \widehat{\delta} \rangle, \quad \Lambda^{A_2} = \langle \varepsilon', \widehat{\delta} \rangle.$$

Both have rank two. Define

$$v_1 := \varepsilon, \quad v_2 := -\varepsilon', \quad \ell_j := \gamma(v_j),$$

where $\gamma : \Lambda \rightarrow \mathbb{Z}$ extracts the $\widehat{\gamma}$ coefficient. Hence

$$(\ell_1, \ell_2) = (1, -1).$$

A Smith normal form calculation gives

$$\sum_{g \in \langle A_1, A_2 \rangle} \text{Im}(g - I) = \ker \gamma = \langle \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \quad \Lambda_{\langle A_1, A_2 \rangle} \cong \mathbb{Z},$$

Moreover, the smallest $\langle A_1, A_2 \rangle$ -invariant sublattice containing Λ_{tor} is precisely $\ker \gamma$. This is the only Smith-form consequence needed in the analytic construction. The complete integral certificate is recorded at its first topological use in Section 5.3.

3 The $(3, 4, \infty)$ orbifold and the period functions

3.1 Orbifold uniformization

Let $B = \mathbb{P}^1$ have coordinate t , and set

$$p_1 = (t = 0), \quad p_2 = (t = 1), \quad p_0 = (t = \infty), \quad t_1 = t, \quad t_2 = t - 1, \quad t_c = t^{-1}.$$

The hyperbolic orbifold of signature $(0; 3, 4, \infty)$ is uniformized by the upper half-plane $\mathfrak{h}$ [2, Ch. 10]. There is a Fuchsian group

$$\Delta = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle \cong \mathbb{Z}/3 * \mathbb{Z}/4, \quad g_0 = (g_1 g_2)^{-1},$$

and a holomorphic quotient map $\pi : \mathfrak{h} \rightarrow B \setminus \{p_0\}$. The element g_j fixes a point $z_j \in \pi^{-1}(p_j)$, and one can choose a local coordinate s_j such that

$$s_j(z_j) = 0, \quad s_j \circ g_j = e^{-2\pi i/m_j} s_j, \quad t_j \circ \pi = s_j^{m_j}.$$

g_0 is a primitive parabolic element. A sufficiently small cusp neighborhood U_r is $\langle g_0 \rangle$ -invariant and simply connected, and

$$U_r / \langle g_0 \rangle \cong \{0 < |t_c| < r\}.$$

We use the global clockwise convention fixed in Section 1; the relation $g_0 = (g_1 g_2)^{-1}$ then fixes the signs at all three punctures simultaneously.

3.2 The period-function theorem

Theorem 3.1 (Period functions). *There exist holomorphic functions*

$$\tau : \mathfrak{h} \rightarrow \mathfrak{h}, \quad \mu, \beta : \mathfrak{h} \rightarrow \mathbb{C}$$

satisfying

$$j(\tau) = 1728 t \circ \pi, \tag{1}$$

$$\tau \circ g_1 = \frac{\tau - 1}{\tau}, \quad \tau \circ g_2 = -\frac{1}{\tau}, \quad \tau \circ g_0 = \tau - 1, \tag{2}$$

$$\mu \circ g_1 = \frac{1 - \mu}{\tau}, \quad \mu \circ g_2 = 1 + \frac{\mu}{\tau}, \quad \mu \circ g_0 = \mu, \tag{3}$$

$$\beta \circ g_1 = \beta + 2 - \frac{6(1 - \mu)^2}{\tau}, \quad \beta \circ g_2 = \beta - 3 - \frac{6\mu^2}{\tau}, \quad \beta \circ g_0 = \beta + 1. \tag{4}$$

Here $\tau(z_1) = e^{i\pi/3}$ and $\tau(z_2) = i$; the function μ is bounded on U_r , and $\beta + \tau$ is bounded there. After adding a suitable constant to β , one can also arrange that

$$D := \operatorname{Im} \beta - \frac{6(\operatorname{Im} \mu)^2}{\operatorname{Im} \tau} < 0 \quad \text{everywhere on } \mathfrak{h}. \tag{5}$$

Proof. We divide the proof into four steps.

Step 1: construction of τ . The modular function j defines the regular covering

$$\mathfrak{h} \setminus (\operatorname{PSL}_2(\mathbb{Z})\rho \cup \operatorname{PSL}_2(\mathbb{Z})i) \longrightarrow \mathbb{C} \setminus \{0, 1728\}$$

The function $F := 1728 t \circ \pi$ has zeros of order 3 over p_1 , whereas $F - 1728$ has zeros of order 4 over p_2 . Hence the corresponding small meridians map in the deck group to the cube of an element of order three and the fourth power of an element of order two; both are the identity. The covering-lift criterion therefore gives $F = j \circ \tau$. It really is enough here to inspect these small meridians. The image of any loop is compact; fill it by a disk in $\mathfrak{h}$, perturb the disk to be transverse to the discrete set $F^{-1}\{0, 1728\}$, and apply van Kampen to express the original loop as a product of conjugates of finitely many small meridians. Thus every loop has trivial monodromy, not merely the loops near the elliptic points. Near a branch point, map $\mathfrak{h}$ to the unit disk; then τ is bounded and extends by the removable singularity theorem. Comparing local orders shows that $\tau - \rho$ has a simple zero at z_1 , whereas $\tau - i$ has a double zero at z_2 .

Write $\tau \circ g = \phi(g) \circ \tau$. After normalization,

$$\phi(g_1) : \tau \mapsto \frac{\tau - 1}{\tau}.$$

If $\phi(g_2) = 1$, then the expansion of $\tau - \tau(z_2)$ in s_2 would involve only powers in $4\mathbb{Z}$, contradicting its double zero. Thus $\phi(g_2)$ has order two. Put $P = \phi(g_0)$. If $P = 1$, then $q = e^{2\pi i \tau}$ has a single-valued holomorphic logarithm on the punctured disk, while $j = q^{-1}(1 + O(q))$ says that q has a positive-order zero at the origin. Its winding number would be both zero and positive, a contradiction. For $|j|$ sufficiently large, every component of $\{|j| > R\}$ lies between two horodisks based at the same cusp. Since P preserves one such component, it fixes a cusp and is parabolic. Write

$$P = I + k \begin{pmatrix} -pr & p^2 \\ -r^2 & pr \end{pmatrix}, \quad \gcd(p, r) = 1, \quad k \neq 0.$$

Because $\phi(g_2) = \phi(g_1)^{-1}P^{-1}$ has trace zero, $-k(p^2 - pr + r^2) = 1$. Positive definiteness forces $k = -1$ and $(p, r) = \pm(1, 0), \pm(0, 1), \pm(1, 1)$. Conjugating by $\phi(g_1)$, we may take $P : \tau \mapsto \tau - 1$, and then $\phi(g_2) : \tau \mapsto -1/\tau$. Only existence is needed, so this normalization is fixed from now on.

On U_r , the function $q = e^{2\pi i \tau}$ is g_0 -invariant and hence is a function of t_c . The expansion $j = q^{-1}(1 + O(q)) = 1728/t_c$ gives

$$q = t_c u_1(t_c), \tag{6}$$

where u_1 is holomorphic and nonvanishing near 0. Set

$$h = (2\pi i)^{-1} \log u_1, \quad s = \tau - h(t_c \circ \pi),$$

Then

$$e^{2\pi i s} = t_c \circ \pi, \quad s \circ g_0 = s - 1, \tag{7}$$

and s maps U_r biholomorphically onto a half-plane.

Step 2: construction of μ . Define the factors of automorphy

$$\tilde{j}(g_1, z) = -\tau(z), \quad \tilde{j}(g_2, z) = \tau(z),$$

and extend them by the cocycle rule. Direct multiplication using (2) shows that the factors for g_1^3 and g_2^4 are 1, and that $\tilde{j}(g_0) = 1$. Let $\mathcal{L}$ be the sheaf of local holomorphic functions satisfying

$$\nu \circ g = \nu / \tilde{j}(g, \cdot)$$

and regular on the chosen cusp branch. We now compute this line bundle, including its elliptic characters. The form E_6 has a simple zero at i , whereas $\tau - i$ has a double zero at z_2 . Thus the divisor of $E_6 \circ \tau$ on the simply connected surface $\mathfrak{h}$ is even, and there is a holomorphic function

$$H = (E_6 \circ \tau)^{1/2},$$

unique up to an overall sign. It has a simple zero at z_2 and no zero at z_1 . For each of the two transformations in (2), the modular factor $c\tau + d$ equals τ . The weight-six transformation law of E_6 therefore gives

$$H \circ g_j = \chi_j \tau^3 H, \quad \chi_j \in \{\pm 1\}.$$

At z_1 , H is nonzero and $\rho^3 = -1$, so evaluation at the fixed point forces $\chi_1 = -1$. At z_2 , the simple zero of H transforms by $s_2 \mapsto e^{-2\pi i/4} s_2 = -is_2$, while $i^3 = -i$; hence $\chi_2 = 1$.

Now set

$$F_{\text{mod}} := \frac{(E_4^2 \circ \tau) H}{\Delta_{\text{mod}} \circ \tau}.$$

The weights of the three factors are 8, 3, 12, so the preceding characters give exactly

$$F_{\text{mod}} \circ g_1 = -\frac{F_{\text{mod}}}{\tau} = \frac{F_{\text{mod}}}{\tilde{j}(g_1, \cdot)}, \quad F_{\text{mod}} \circ g_2 = \frac{F_{\text{mod}}}{\tau} = \frac{F_{\text{mod}}}{\tilde{j}(g_2, \cdot)}.$$

Thus F_{mod} is a meromorphic section of $\mathcal{L}$, not merely a function with the correct weight.

It remains to distinguish orbifold-forced vanishing from the divisor on the coarse curve. Since E_4 has a simple zero at ρ and $\tau - \rho$ has a simple zero at z_1 , the section F_{mod} has order 2 in s_1 . The homogeneous factor at z_1 is $-1/\rho = e^{2\pi i/3} = (e^{-2\pi i/3})^2$, so every local generator of $\mathcal{L}$ has exactly this forced order modulo 3. Choose an equivariant local frame with leading term s_1^2 ; the quotient of F_{mod} by that frame is an invariant unit. At z_2 , H has order 1, and the homogeneous factor is $1/i = -i = e^{-2\pi i/4}$. An equivariant local frame there has leading term s_2 , and the same quotient argument gives an invariant unit. Consequently neither elliptic zero contributes to the divisor of the descended line bundle.

At the cusp, E_4 and E_6 are units and $\Delta_{\text{mod}} = q(1 + O(q))$. Since $q = t_c u_1(t_c)$, the section F_{mod} is t_c^{-1} times a unit. Its coarse divisor is therefore exactly $-p_0$, and $\mathcal{L} \cong \mathcal{O}_{\mathbb{P}^1}(-p_0)$. The local solutions of (3) form an $\mathcal{L}$ -torsor: the difference of two solutions obeys the homogeneous factor law, and adding a homogeneous local section to one solution produces another. Explicit local solutions are 0 on a chosen branch at an ordinary point, 0 on U_τ at the cusp, and, near z_1, z_2 , respectively,

$$\mu_1 = \frac{2 - \tau}{3}, \quad \mu_2 = \frac{1 - \tau}{2}.$$

Since $H^1(\mathbb{P}^1, \mathcal{O}(-1)) = 0$, the torsor obstruction vanishes and a global μ exists. It is bounded at the cusp because the chosen cusp representative is 0 and the transition sections of $\mathcal{L}$ are regular there.

Step 3: construction of β . Set

$$\varphi_1 = 2 - \frac{6(1 - \mu)^2}{\tau}, \quad \varphi_2 = -3 - \frac{6\mu^2}{\tau}.$$

Repeated substitution of (2) and (3) gives

$$\sum_{k=0}^2 \varphi_1 \circ g_1^k = 0, \quad \sum_{k=0}^3 \varphi_2 \circ g_2^k = 0.$$

Thus the local equation $\beta \circ g_j - \beta = \varphi_j$ is compatible with the finite-order relations. Take 0 on a local branch at an ordinary point, take $\beta = -\tau$ at the cusp, and near z_j take

$$\beta = \frac{1}{m_j} \sum_{k=0}^{m_j-1} k(\varphi_j \circ g_j^k).$$

These local solutions form an $\mathcal{O}_B$ -torsor: on an overlap, the difference of two solutions is an orbifold-invariant holomorphic function and therefore descends to B . Its obstruction lies in $H^1(\mathbb{P}^1, \mathcal{O}) = 0$, so a global solution exists; two global solutions differ by a constant in $H^0(\mathbb{P}^1, \mathcal{O}) = \mathbb{C}$.

Step 4: nondegeneracy. Since $b := \beta + \tau$ and μ are bounded at the cusp,

$$D = \text{Im } b - \text{Im } \tau - \frac{6(\text{Im } \mu)^2}{\text{Im } \tau} \longrightarrow -\infty.$$

The invariance of D follows from the explicit identity

$$\Im\left(-\frac{6a^2}{\tau}\right) - \frac{6\Im(a/\tau)^2}{\Im(-1/\tau)} = -\frac{6(\Im a)^2}{\Im \tau} \quad (\tau \in \mathfrak{h}, a \in \mathbb{C}). \quad (8)$$

To check it, write $\tau = x + iy$ and $a = r + is$ and multiply by $|\tau|^2 y$; both sides become $-6s^2|\tau|^2$. Apply (8) with $a = \mu$ for g_2 and with $a = 1 - \mu$ for g_1 . The remaining constants in (3)–(4) are real, while $\Im(1 - \mu) = -\Im\mu$. Therefore $D(g_j z) = D(z)$ for both generators, so D is Δ -invariant. It is finite and continuous at p_1, p_2 , so it has an upper bound M on the compact complement of a small cusp neighborhood. Adding a constant c_0 to β adds $\Im c_0$ to D ; choosing $\Im c_0 < -M$ yields (5). $\square$

3.3 The period matrix, its lattice, and equivariance

Define

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix} = [Z(z) \mid I_2], \quad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}. \quad (9)$$

Regard Π as a real-linear map $\Lambda \otimes \mathbb{R} \rightarrow \mathbb{C}^2$. Expanding along the last two columns gives

$$\det_{\mathbb{R}} \Pi = -\det \operatorname{Im} Z = \operatorname{Im} \tau \operatorname{Im} \beta - 6(\operatorname{Im} \mu)^2 = \operatorname{Im} \tau D < 0. \quad (10)$$

Thus $L(z) := \Pi(z)\Lambda$ is a full lattice in $\mathbb{C}^2$.

Set

$$\sigma_1 = 6\mu\gamma + \tau u + w, \quad \sigma_2 = \beta\gamma + \mu u + \delta.$$

Direct substitution of the matrices T_j and the three transformation laws gives

$$\begin{aligned} \sigma_1 \circ g_1 &= -\tau^{-1} T_1 \sigma_1, & \sigma_2 \circ g_1 &= T_1 \sigma_2 + (1 - \mu) \tau^{-1} T_1 \sigma_1, \\ \sigma_1 \circ g_2 &= \tau^{-1} T_2 \sigma_1, & \sigma_2 \circ g_2 &= T_2 \sigma_2 - \mu \tau^{-1} T_2 \sigma_1, \\ \sigma_i \circ g_0 &= T_0 \sigma_i. \end{aligned}$$

Consequently there are invertible matrices

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1 - \mu)/\tau & 1 \end{pmatrix}, \quad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix}, \quad R_{g_0} = I,$$

satisfying

$$\Pi(gz) = R_g(z) \Pi(z) A_g^{-1}, \quad R_{gh}(z) = R_g(hz) R_h(z). \quad (11)$$

Here $A_{g_1} = A_1$, $A_{g_2} = A_2$, $A_{g_0} = M_0$, and A_g for a word g is the corresponding homology monodromy. The inverse is forced by duality: the displayed transformations of the cohomology rows give $\Pi(gz) = R_g \Pi T_g^t$, while $A_g = (T_g^{-1})^t$. Equivalently,

$$R_g(z) \Pi(z) = \Pi(gz) A_g. \quad (12)$$

This last form shows directly that the flat homology coordinate transforms by $x \mapsto A_g x$.

Let Λ act on $\mathfrak{h} \times \mathbb{C}^2$ by

$$\lambda \cdot (z, \zeta) = (z, \zeta + \Pi(z)\lambda),$$

and let g act by $(z, \zeta) \mapsto (gz, R_g(z)\zeta)$. Equation (11) says that the latter action normalizes the former. Hence

$$\mathcal{T} = (\mathfrak{h} \times \mathbb{C}^2) / \Lambda$$

is a family of complex two-tori over $\mathfrak{h}$. Remove $\Delta_{z_1} \cup \Delta_{z_2}$ and then quotient by Δ to obtain

$$f^\circ : J \longrightarrow B^\circ := B \setminus \{p_0, p_1, p_2\}, \quad (13)$$

It is a proper holomorphic submersion with a holomorphic zero section. The clockwise meridians act on H_1 by A_1, A_2, M_0 , respectively.

At the cusp, $\tau = s + h$ and $\beta = b - s - h$ give

$$Z(z) = s(z)B_0 + C(t_c), \quad C(t_c) = \begin{pmatrix} 6\mu(t_c) & h(t_c) \\ b(t_c) - h(t_c) & \mu(t_c) \end{pmatrix}, \quad (14)$$

where C is holomorphic at $t_c = 0$. Therefore, for $\bar{\lambda} \in \bar{\Lambda}$,

$$\exp(2\pi i Z \bar{\lambda}) = t_c^{B_0 \bar{\lambda}} c_{\bar{\lambda}}(t_c), \quad c_{\bar{\lambda}}(t_c) := \exp(2\pi i C(t_c) \bar{\lambda}). \quad (15)$$

4 Compactification

4.1 The cusp p_0 : the infinite-fan toroidal filling

This section uses (14)–(15) to construct a proper complex threefold near the cusp.

4.1.1 The A_2 triangulation and the infinite fan

Let $N' = \mathbb{Z}^3$, write its elements as (y, y_3) with $y \in \mathbb{Z}^2$, and put $M' = \text{Hom}(N', \mathbb{Z})$. On the algebraic torus

$$T_{N'} = (\mathbb{C}^*)^3$$

use coordinates (x_1, x_2, t_c) . Let $\mathcal{T}_{A_2}$ be the periodic triangulation of $\mathbb{R}^2$ with vertex set $\mathbb{Z}^2$ and triangles

$$\{v, v + e_1, v + e_2\}, \quad \{v + e_1, v + e_2, v + e_1 + e_2\} \quad (v \in \mathbb{Z}^2).$$

The edge directions are $\pm e_1, \pm e_2, \pm(e_1 - e_2)$. For every cell P , put

$$\hat{P} = \text{cone}(P \times \{1\}) \subset N'_{\mathbb{R}}, \quad \mathcal{F} = \{0\} \cup \{\hat{P}\}.$$

Lemma 4.1 (Local model of the fan). *$\mathcal{F}$ is a locally finite smooth rational fan with support $(\mathbb{R}^2 \times \mathbb{R}_{>0}) \cup \{0\}$. The associated toric manifold $Y = Y_{\mathcal{F}}$ is a connected, Hausdorff, second-countable smooth complex threefold. The function t_c extends holomorphically to Y , and*

$$t_c^{-1}(0) = \bigcup_{v \in \mathbb{Z}^2} E_v, \quad \text{div}(t_c) = \sum_v E_v.$$

If a triangle σ has vertices v_0, v_1, v_2 , with dual basis $m_0, m_1, m_2 \in M'$, then

$$U_{\sigma} \cong \mathbb{C}^3, \quad z_i = \chi^{m_i}, \quad t_c = z_0 z_1 z_2, \quad E_{v_i} \cap U_{\sigma} = (z_i = 0).$$

Each $E_v \cong dP_6$, and its toric boundary is an anticanonical hexagon of six (-1) -curves.

Proof. The ray generators for the two kinds of triangles are, respectively,

$$((v, 1), (v + e_1, 1), (v + e_2, 1)), \quad ((v + e_1, 1), (v + e_2, 1), (v + e_1 + e_2, 1)).$$

The determinants are 1 or -1 , so every cone is unimodular. Intersections of triangles are common faces, hence intersections of their cones are common faces; local finiteness follows from $\mathcal{T}_{A_2}$. For every three-dimensional cone σ , take the standard affine toric chart $U_{\sigma} = \text{Spec } \mathbb{C}[\sigma^{\vee} \cap M']$. The common-face property glues U_{σ} and $U_{\sigma'}$ along the same toric open set. Unimodularity gives $U_{\sigma} \cong \mathbb{C}^3$, so the resulting space is smooth. Any two points involve only finitely many adjacent cones. Place these cones in a finite subfan and apply the usual separatedness argument for a finite fan [5, §1.4]; it produces disjoint neighborhoods, so the infinite gluing is still Hausdorff. The cones are indexed by $v \in \mathbb{Z}^2$, hence the atlas is countable and the space is second countable.

Thus the infinite fan loses quasi-compactness but none of the local properties required of a complex manifold.

The height function $(0, 0, 1) \in M'$ takes value 1 on every primitive ray $(v, 1)$, so t_c vanishes to order exactly one along each E_v . In a unimodular three-cone chart, $(0, 0, 1) = m_0 + m_1 + m_2$, hence $t_c = z_0 z_1 z_2$ and the central fiber is reduced with normal crossings. The star fan of E_v in $N'/\mathbb{Z}(v, 1)$ has the cyclic rays

$$e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2$$

which form exactly the hexagonal fan of dP_6 . $\square$

4.1.2 The lattice action

For $\bar{\lambda} \in \bar{\Lambda} \cong \mathbb{Z}^2$, define

$$\phi_{\bar{\lambda}}(y, y_3) = (y + y_3 B_0 \bar{\lambda}, y_3).$$

On the height-one plane this is an integral translation, so it preserves $\mathcal{F}$ and induces a toric automorphism $\Phi_{\bar{\lambda}}$. On the dense torus,

$$\Phi_{\bar{\lambda}}(x, t_c) = (t_c^{B_0 \bar{\lambda}} x, t_c).$$

Correct it by the unit in (15) and define

$$\Psi_{\bar{\lambda}}(p) = (c_{\bar{\lambda}}(t_c(p)), 1) \cdot \Phi_{\bar{\lambda}}(p). \quad (16)$$

Since $c_{\bar{\lambda}} c_{\bar{\mu}} = c_{\bar{\lambda} + \bar{\mu}}$ and $\Phi_{\bar{\lambda}}$ acts trivially on the (x_1, x_2) subtorus, $\Psi_{\bar{\lambda}} \Psi_{\bar{\mu}} = \Psi_{\bar{\lambda} + \bar{\mu}}$.

Theorem 4.2 (The cusp quotient). *For sufficiently small $\epsilon > 0$, the action (16) of $\bar{\Lambda}$ on*

$$Y_{\epsilon} := \{|t_c| < \epsilon\} \subset Y$$

is free and properly discontinuous. The quotient

$$N_0 := Y_{\epsilon} / \bar{\Lambda}$$

is a Hausdorff complex threefold, and t_c descends to a proper surjection

$$f_0 : N_0 \rightarrow \Delta_{\epsilon}.$$

For $t_c \neq 0$, f_0 is a submersion by complex two-tori. The central fiber $W = f_0^{-1}(0)$ is reduced, compact, irreducible, and has normal crossings.

Proof. Put

$$R(t_c) = -2\pi \operatorname{Im} C(t_c), \quad B_t = B_0 + \frac{R(t_c)}{\log |t_c|}.$$

Choose ϵ so that $|\log \epsilon| \geq 2 \sup_{|t_c| \leq r/2} \|R(t_c)\|$. Then

$$\|B_t - B_0\| \leq \frac{1}{2}, \quad |B_t \bar{\lambda}| \geq \frac{1}{2} |\bar{\lambda}|.$$

Freeness. On the dense torus, if $\Psi_{\bar{\lambda}}(x, t_c) = (x, t_c)$, taking logarithms of absolute values gives

$$\log |t_c| B_0 \bar{\lambda} + R(t_c) \bar{\lambda} = 0.$$

The norm of the first term is at least $|\log |t_c|| |\bar{\lambda}|$, while the norm of the second is at most $\|R\|_{\infty} |\bar{\lambda}|$. This is impossible unless $\bar{\lambda} = 0$. On the central fiber, $\Psi_{\bar{\lambda}}$ sends the cell orbit $O_{\hat{P}}$ to $O_{\widehat{P+B_0 \bar{\lambda}}}$. Since P is bounded and B_0 is unimodular, a nonzero translation cannot fix it.

Proper discontinuity. For a torus point, define

$$y(p) = \frac{\log |x(p)|}{\log |t_c(p)|} \in \mathbb{R}^2.$$

If $U_\sigma(2) = \{|z_i| < 2\}$, the relation $\log |z_i| = \log |t_c| \theta_i^\sigma(y)$ shows that, when $p \in U_\sigma(2)$, the point $y(p)$ lies in a fixed bounded enlargement of σ . Moreover,

$$y(\Psi_{\bar{\lambda}} p) = y(p) + B_t \bar{\lambda}. \quad (17)$$

Therefore, if $\Psi_{\bar{\lambda}} U_\sigma(2) \cap U_{\sigma'}(2) \neq \emptyset$, $|\bar{\lambda}|$ is bounded by a constant depending only on σ, σ' . A compact set is covered by finitely many charts, so on the part $t_c \neq 0$ only finitely many $\bar{\lambda}$ can make it intersect one of its translates. On $t_c = 0$, an intersection of charts requires the translated cell $P + B_0 \bar{\lambda}$ to share a face with a fixed cell P' . Local finiteness of the triangulation and unimodularity of B_0 again leave only finitely many $\bar{\lambda}$. Combining the two parts proves proper discontinuity on all of Y_ϵ .

The quotient by a free properly discontinuous action is a complex manifold and the projection is a covering. The Hausdorff property can also be checked directly: compact neighborhoods of distinct orbits meet only finitely many translates, and shrinking them one at a time separates their quotient images.

Properness. Fix $0 < \epsilon_1 < \epsilon$. By (17) and invertibility of B_t , for every point with $t_c \neq 0$ one can choose $\bar{\lambda} \in \mathbb{Z}^2$ so that $B_t^{-1}y + \bar{\lambda} \in [-1/2, 1/2]^2$. By (17), the translated point has

$$y' = y + B_t \bar{\lambda} = B_t(B_t^{-1}y + \bar{\lambda}),$$

which lies in one fixed disk because B_t is uniformly bounded. Choose a triangle σ containing y' and write $y' = \sum_i \theta_i v_i$ in its barycentric coordinates. Then $\theta_i \geq 0$, $\sum_i \theta_i = 1$, and the toric coordinates satisfy

$$|z_i| = |t_c|^{\theta_i} \leq 1, \quad t_c = z_0 z_1 z_2.$$

Only finitely many triangles meet the fixed disk. Thus every orbit with $0 < |t_c| \leq \epsilon_1$ meets the union of the unit polydisks in finitely many charts.

For $t_c = 0$, let the point lie in the toric orbit indexed by a cell P , choose a vertex $v_0 \in P$, and solve $B_0 \bar{\lambda} = -v_0$. The translated cell $P - v_0$ contains 0, so its cone contains the ray $\mathbb{R}_{\geq 0}(0, 1)$. The translated orbit therefore lies in the corresponding toric divisor $E_0 \cong dP_6$. Hence the compact surface E_0 meets every $\bar{\lambda}$ -orbit in the central fiber; this also controls the residual $(\mathbb{C}^*)^d$ coordinates along each toric orbit, which cell normalization alone would not control.

For each of the finitely many charts needed over $0 < |t_c| \leq \epsilon_1$, set

$$K_\sigma = \{p \in U_\sigma : |z_0|, |z_1|, |z_2| \leq 2, |t_c(p)| = |z_0 z_1 z_2| \leq \epsilon_1\}.$$

Each K_σ is a closed subset of a compact polydisk and lies in Y_ϵ . The finite union $K = E_0 \cup \bigcup K_\sigma$ is therefore compact and meets every orbit over $|t_c| \leq \epsilon_1$. Consequently $f_0^{-1}(|t_c| \leq \epsilon_1)$ is the continuous image of the compact set K in the quotient and is compact. This holds for every $\epsilon_1 < \epsilon$, so f_0 is proper.

Over $t_c = t_0 \neq 0$, the exponential covering $\mathbb{C}^2 \rightarrow (\mathbb{C}^*)^2$ gives

$$f_0^{-1}(t_0) \cong \mathbb{C}^2 / (\mathbb{Z}^2 + Z(s)\mathbb{Z}^2) = \mathbb{C}^2 / L(z),$$

where $e^{2\pi i s} = t_0$. This is a compact torus, also by (10). In the covering space the central fiber is defined by $t_c = z_0 z_1 z_2$, so it is reduced with normal crossings. The covering projection is locally biholomorphic, hence these properties descend. Compactness follows from properness; connectedness and irreducibility are proved next. $\square$

4.1.3 The central fiber and its normalization

Proposition 4.3. *The normalization map*

$$\nu : E_0 \cong dP_6 \longrightarrow W$$

is finite and surjective, and is injective away from the hexagonal boundary. It identifies the three pairs of opposite sides of the hexagon. The double locus consists of three copies of $\mathbb{P}^1$, meeting exactly at two triple points P, Q . In particular,

W is irreducible and connected.

Proof. $B_0\bar{\Lambda} = \mathbb{Z}^2$, so the action on vertices is simply transitive. Every cell has a vertex; therefore every torus orbit in the central fiber is equivalent to an orbit in E_0 , proving surjectivity. Two points of E_0 can be identified only when they arise from cells P, P' both containing 0 and $P' = P + w$. If P is an edge, the only possibility is

$$P = \{0, v\}, \quad P' = \{0, -v\}, \quad w = -v;$$

If P is a triangle, the six triangles containing 0 split into upper and lower classes, which contract to P and Q . Hence ν is injective on the open torus and has finite fibers. Since E_0 is compact, ν is proper; a proper holomorphic map with finite fibers is finite. It is bimeromorphic from the normal surface E_0 to the reduced surface W , so it is the normalization. The three unoriented edge directions give the three double curves, each passing through the upper and lower triangle classes, hence through exactly P, Q . $\square$

4.1.4 Biholomorphic identification with the punctured family

Set

$$U_\epsilon := \{z \in U_r : 0 < |t_c \circ \pi(z)| < \epsilon\}.$$

Define

$$E_0^{\text{exp}} : U_\epsilon \times \mathbb{C}^2 \rightarrow (\mathbb{C}^*)^3, \quad (z, \zeta) \mapsto (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}).$$

Its fibers are generated by translations in Λ_{tor} and by g_0 . For $\lambda \in \Lambda$, equation (15) gives

$$E_0^{\text{exp}}(z, \zeta + \Pi(z)\lambda) = \Psi_{\bar{\lambda}}(E_0^{\text{exp}}(z, \zeta)).$$

It therefore descends to a holomorphic map on the two quotients. If two source points have the same image, their exponential coordinates differ by an element of Λ_{tor} and their remaining difference is a unique $\bar{\lambda}$ -translation, so they already agree in J ; hence the map is injective. Conversely, every point of Y_ϵ with $t_c \neq 0$ has logarithms of x_1, x_2, t_c , and two choices differ by the same deck transformations, proving surjectivity. The local logarithms also provide the holomorphic inverse. Thus it is the biholomorphism

$$G_0 : J|_{0 < |t_c| < \epsilon} \xrightarrow{\sim} N_0 \setminus W. \quad (18)$$

The zero section corresponds to $(1, 1, t_c)$. On the cone

$$\text{cone}((0, 1), (e_1, 1), (e_2, 1)),$$

use coordinates $(t_c/(x_1 x_2), x_1, x_2)$. The section extends to the closed disk $(t_c, 1, 1)$, which meets W transversely in its smooth open orbit.

4.2 The elliptic points p_1, p_2 : logarithmic transforms

Fix $j \in \{1, 2\}$. Choose a g_j -invariant simply connected disk $\Delta_j \ni z_j$ and a linearizing coordinate s_j satisfying

$$s_j \circ g_j = \zeta_j s_j, \quad \zeta_j = e^{-2\pi i/m_j}, \quad t_j \circ \pi = s_j^{m_j}.$$

Let $J'_j = \mathcal{T}|_{\Delta_j}$. For $b \in \Lambda \otimes \mathbb{R}$, define

$$\tilde{g}_j^b(z, \zeta) = (g_j z, R_{g_j}(z)\zeta + \Pi(g_j z)b).$$

Because $R_{g_j}\Pi = \Pi \circ g_j$, this descends to a biholomorphic automorphism of J'_j . In the real flat coordinate $x \in (\Lambda \otimes \mathbb{R})/\Lambda$, it is

$$(z, x) \mapsto (g_j z, A_j x + b).$$

For the chosen $v_j \in \Lambda^{A_j}$, set

$$\tilde{g}_j^{\log} := \tilde{g}_j^{v_j/m_j}. \quad (19)$$

Lemma 4.4 (Freeness of the chosen actions). *For $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$, the group $\langle \tilde{g}_j^{\log} \rangle \cong \mathbb{Z}/m_j$ acts freely on J'_j .*

Proof. Since $A_j v_j = v_j$, the k th power on the central torus is $x \mapsto A_j^k x + k v_j/m_j$. Its m_j th power is the lattice translation t_{v_j} , hence the identity on the torus. A fixed point of a nontrivial power must lie over $s_j = 0$. A finite-order affine map $x \mapsto Ax + b$ of $(\Lambda \otimes \mathbb{R})/\Lambda$ has a fixed point if and only if every A -invariant integral linear form $\phi \in V$ satisfies $\phi(b) \in \mathbb{Z}$. Indeed, the fixed-point equation is $(A - I)x \equiv -b \pmod{\Lambda}$. Finite order gives the decomposition $\Lambda \otimes \mathbb{R} = \ker(A - I) \oplus \text{Im}(A - I)$. Projecting the equation to the fixed subspace says that the image of b in the fixed torus must vanish. The integral dual of that torus is exactly V^A , so this is equivalent to the integrality of $\phi(b)$ for every $\phi \in V^A$. Conversely, that condition kills the fixed component, while $A - I$ is invertible on the second summand, where one can solve for x .

Here

$$V^{T_1} = \langle \gamma, 2u + w + 3\delta \rangle, \quad V^{T_2} = \langle \gamma, u + w + 2\delta \rangle.$$

In particular, γ is invariant under every power of either action. For $k = 1, 2$, the translation in the k th power of the first action is $k v_1/3$, and

$$\gamma(k v_1/3) = k/3 \notin \mathbb{Z}.$$

For the second action, the powers $k = 1, 3$ have $\gamma(k v_2/4) = -k/4 \notin \mathbb{Z}$, while the square has $\gamma(v_2/2) = -1/2 \notin \mathbb{Z}$. In every nontrivial power, the invariant form γ therefore violates the necessary fixed-point integrality condition. Hence both actions are free. $\square$

Define

$$N_j := J'_j / \langle \tilde{g}_j^{\log} \rangle, \quad f_j : N_j \rightarrow D_j := \Delta_j / \langle g_j \rangle.$$

The function $s_j^{m_j}$ descends to t_j , and

$$f_j^*(p_j) = m_j S_j, \quad S_j = (\mathbb{C}^2/L(z_j)) / \langle x \mapsto A_j x + v_j/m_j \rangle.$$

The free finite quotient makes N_j smooth, and S_j is a smooth compact complex surface.

To glue this model to J , set

$$\sigma_j(z) = \frac{\log s_j(z)}{2\pi i} \Pi(z) v_j \quad \text{as a section of } J'_j|_{\Delta_j^*}. \quad (20)$$

Different logarithm branches differ by $k\Pi v_j$ and hence define the same point of the torus. Since

$$\log s_j \circ g_j = \log s_j - 2\pi i/m_j, \quad A_j v_j = v_j$$

let h_j be fiberwise translation by σ_j . The extra translation term after conjugation is

$$-R_{g_j}(z)\sigma_j(z) + \Pi(g_j z) \frac{v_j}{m_j} + \sigma_j(g_j z).$$

The identities $\log s_j \circ g_j = \log s_j - 2\pi i/m_j$, $R_{g_j}\Pi = \Pi \circ g_j A_j$, and $A_j v_j = v_j$ make this expression zero. Therefore

$$h_j \tilde{g}_j^{\log} h_j^{-1} = \tilde{g}_j^0.$$

This yields the biholomorphic identification

$$G_j : N_j \setminus S_j \xrightarrow{\sim} J|_{D_j^*}. \quad (21)$$

The affine action also records the fundamental group needed below:

$$\Gamma_j = \langle \Lambda, \hat{g}_j \mid \hat{g}_j \lambda \hat{g}_j^{-1} = A_j \lambda, \hat{g}_j^{m_j} = t_{v_j} \rangle. \quad (22)$$

4.3 The global compact complex threefold

Choose pairwise disjoint small disks D_0, D_1, D_2 and use (18), (21) to glue

$$X = (J \sqcup N_0 \sqcup N_1 \sqcup N_2) / \sim. \quad (23)$$

All transition maps are biholomorphic and preserve the base coordinate, so they define a holomorphic map $f : X \rightarrow B$.

Theorem 4.5. *X is a connected compact Hausdorff complex threefold, and $f : X \rightarrow \mathbb{P}^1$ is a proper surjection with connected fibers. Over B° the fibers are complex two-tori; $f^{-1}(p_0) = W$ is reduced with normal crossings; $f^*(p_j) = m_j S_j$.*

Proof. Because the three disks are disjoint, every nontrivial equivalence class consists of one point in an annular overlap and its image in the corresponding filling. Saturations of open sets are open, so the four pieces map to X as open embeddings. The transition functions of the complex atlas are precisely $G_0^{\pm 1}, G_j^{\pm 1}$; hence X is a complex manifold.

If two points have different images in B , inverse images of disjoint disks separate them. If their images agree, the two points lie together in a Hausdorff chart $f^{-1}(B^\circ)$ or $f^{-1}(D_i)$. Thus X is Hausdorff. On the displayed open cover, f equals f°, f_0, f_j . The torus bundle f° is locally a quotient of a product by a continuously varying full lattice, so over a sufficiently small closed base disk it is a compact torus bundle and hence proper. The map f_0 is proper by Theorem 4.2; each f_j is the finite quotient of the proper torus family $J'_j \rightarrow \Delta_j$, hence is proper. Properness of a map between locally compact Hausdorff spaces can be checked locally on the base: for compact $K \subset B$, choose finitely many relatively compact small neighborhoods U_α covering K such that every $f^{-1}(\overline{U}_\alpha)$ is compact. Then $f^{-1}(K)$ is closed in a finite union of these compact sets, so it is compact. Hence f is proper. Since B is compact, $X = f^{-1}(B)$ is compact. The spaces J, N_0, N_j are connected and meet along the annular overlaps, so X is connected. A general fiber is a torus; S_j is a finite quotient of a connected torus; and W is connected by Proposition 4.3. Thus every fiber is connected. $\square$

5 Integral topology of the fillings

5.1 The cusp filling

The toric covering space Y_ϵ is simply connected. Indeed, choose a cofinal exhaustion by connected finite subfans, every one of which contains the three rays $(0, 1), (e_1, 1), (e_2, 1)$. The fundamental group of each finite smooth toric variety is N' modulo its ray generators; these three rays already generate N' . Thus every finite stage is simply connected.

It remains to keep a null-homotopy inside the sublevel set. Let a loop $\alpha \subset Y_\epsilon$ lie in a finite stage $Y_{\mathcal{F}'}$, choose a null-homotopy $H : D^2 \rightarrow Y_{\mathcal{F}'}$, and put $M = \max_{D^2} |t_c \circ H|$. The torus element $a_r = (1, 1, r) \in (\mathbb{C}^*)^3$, $0 < r \leq 1$, acts on every toric stage and satisfies $t_c(a_r p) = r t_c(p)$. Choose $r_0 > 0$ with $r_0 M < \epsilon$. Then $a_{r_0} \circ H$ contracts the scaled loop $a_{r_0} \circ \alpha$ inside Y_ϵ , while $(s, u) \mapsto a_{(1-s)+sr_0} \alpha(u)$ is a homotopy in Y_ϵ from α to the scaled loop. Concatenating this annulus with the scaled disk contracts α in Y_ϵ . Since $Y_\epsilon \rightarrow N_0$ is the covering with deck group $\bar{\Lambda}$,

$$\pi_1(N_0) = \bar{\Lambda} \cong \mathbb{Z}^2. \quad (24)$$

The map from the first homology Λ of a general fiber to $\pi_1(N_0)$ is the natural projection. Consequently Λ_{tor} and the cusp meridian of the extended zero section vanish in N_0 .

5.2 Boundary data and the fundamental group of X

For the chosen gluing, the cusp identification G_0 is untranslated, while the elliptic translations are $v_1 = \varepsilon$ and $v_2 = -\varepsilon'$. Thus, writing $\ell_0 = 0$ and $\ell_j = \gamma(v_j)$,

$$(\ell_0, \ell_1, \ell_2) = (0, 1, -1). \quad (25)$$

Changing a branch of $\log s_j$ adds the lattice period Πv_j , so these torus-valued gluing sections are unambiguous.

Let a_1, a_2 be the clockwise meridians in B° , with $a_0 = (a_1 a_2)^{-1}$. Lift them along the zero section to $\hat{\rho}_1, \hat{\rho}_2$. Then

$$\pi_1(J) = \Lambda \rtimes F(\hat{\rho}_1, \hat{\rho}_2), \quad \hat{\rho}_j \lambda \hat{\rho}_j^{-1} = A_j \lambda.$$

Inside N_j , equation (22) gives the relation

$$\hat{\rho}_j^{m_j} = t_{v_j}. \quad (26)$$

Here is the sign check once and for all. Each g_j multiplies s_j by $e^{-2\pi i/m_j}$, the affine translation is $+v_j/m_j$, and the gluing section is $+(\log s_j)/(2\pi i)\Pi v_j$. The conjugation calculation in (20) therefore makes a clockwise meridian lift to $\tilde{g}_j^{\log}$, whose m_j th power is t_{v_j} . The pair-of-pants relation $g_0 = (g_1 g_2)^{-1}$ forces the two elliptic relations to have this same orientation; they can only be inverted simultaneously. Their fiber coefficients are consequently $\gamma(v_1) = 1$ and $\gamma(v_2) = -1$.

In N_0 , Λ_{tor} and the cusp meridian vanish. Because G_0 is untranslated, the cusp boundary relation is simply $\hat{\rho}_1 \hat{\rho}_2 = 1$. Van Kampen's theorem therefore gives

$$\pi_1(X) = \left\langle \Lambda, \hat{\rho}_1, \hat{\rho}_2 \left| \begin{array}{l} \hat{\rho}_j \lambda \hat{\rho}_j^{-1} = A_j \lambda, \\ \hat{\rho}_1^3 = t_{v_1}, \quad \hat{\rho}_2^4 = t_{v_2}, \\ \hat{\rho}_1 \hat{\rho}_2 = 1, \quad \Lambda_{\text{tor}} = 1 \end{array} \right. \right\rangle.$$

The Smith calculation in Section 2 says that the smallest $\langle A_1, A_2 \rangle$ -invariant sublattice containing Λ_{tor} is $\ker \gamma$. Thus the image of Λ reduces to one central element c , the image of $\hat{\gamma}$. Putting $x = \hat{\rho}_1$ and $y = \hat{\rho}_2$ gives

Theorem 5.1.

$$\pi_1(X) = \langle c, x, y \mid c \text{ central}, xy = 1, x^3 = c, y^4 = c^{-1} \rangle = 1.$$

Indeed, $y = x^{-1}$ and $y^4 = c^{-1}$ give $x^4 = c$; together with $x^3 = c$ this gives $x = 1$, and then $y = c = 1$.

5.3 Elliptic fibers and the integral lattice ledger

Let F be a general fiber. Then

$$H^q(F; \mathbb{Z}) = \Lambda^q V.$$

Let $\pi_j : F \rightarrow S_j$ be the unramified covering of degree m_j , and set

$$\eta_1 = 2u + w + 3\delta, \quad \eta_2 = u + w + 2\delta, \quad q = u \wedge w + 6\gamma \wedge \delta.$$

A sublattice is called *saturated* when its quotient in the ambient lattice is torsion-free.

5.3.1 The integral arithmetic ledger

Dually to the fixed lattices in Section 2,

$$V^{T_1} = \langle \gamma, \eta_1 \rangle, \quad V^{T_2} = \langle \gamma, \eta_2 \rangle, \quad V^{T_0} = \langle \gamma, u \rangle, \quad V^{\langle T_1, T_2 \rangle} = \mathbb{Z}\gamma.$$

A direct exterior-algebra calculation gives

$$\begin{aligned} (\Lambda^2 V)^{T_0} &= \langle \gamma u, \gamma \delta, uw, \gamma w - u\delta \rangle, \\ (\Lambda^2 V)^{T_j} &= \langle \gamma \wedge \eta_j, q \rangle, \\ (\Lambda^2 V)^{\langle T_1, T_2 \rangle} &= \mathbb{Z}q, \quad (\Lambda^3 V)_{\langle T_1, T_2 \rangle} = \mathbb{Z}[uw\delta]. \end{aligned}$$

For the integral certificate, use the ordered basis $(\gamma u, \gamma w, \gamma \delta, uw, u\delta, w\delta)$ of $\Lambda^2 V$ and put

$$C_V = [T_1 - I \mid T_2 - I], \quad C_\Lambda = [A_1 - I \mid A_2 - I], \quad C_2 = [\Lambda^2 T_1 - I \mid \Lambda^2 T_2 - I].$$

Integral Euclidean row and column reduction gives the following Smith data; omitted diagonal entries are zero.

matrix	nonzero Smith factors	image or kernel consequence
C_V	1, 1, 1	$\text{Im } C_V = \langle \gamma, u, w \rangle$
C_Λ	1, 1, 1	$\text{Im } C_\Lambda = \ker \gamma$
C_2	1, 1, 1, 1, 1	$\text{Im } C_2 = \langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle$
$M_0 - I$	1, 1	$\text{rk ker} = 2$
$\Lambda^2 M_0 - I$	1, 1	$\text{rk ker} = 4$
$\Lambda^3 M_0 - I$	1, 1	$\text{rk ker} = 2$

The unit Smith factors prove integrally that every displayed image is saturated. In particular,

$$(T_1 - I)V + (T_2 - I)V = \langle \gamma, u, w \rangle, \quad V_G = \mathbb{Z}\bar{\delta}, \quad (27)$$

$$\text{Im}(\Lambda^2 T_1 - I) + \text{Im}(\Lambda^2 T_2 - I) = \langle \gamma u, \gamma w, u\delta, w\delta, uw - 6\gamma\delta \rangle, \quad (\Lambda^2 V)_G = \mathbb{Z}[\gamma\delta], \quad (28)$$

with

$$[uw] = 6[\gamma\delta], \quad [q] = 12[\gamma\delta], \quad (\Lambda^3 V)_G = \mathbb{Z}[uw\delta].$$

The same Smith data show that the cusp invariant lattices have ranks 1, 2, 4, 2, 1 in degrees 0, ..., 4, with no hidden torsion.

5.3.2 Integral homology of the elliptic fibers

Abelianizing (22) gives

$$H_1(S_j; \mathbb{Z}) = (\Lambda / (A_j - I)\Lambda \oplus \mathbb{Z}\hat{g}_j) / ([v_j], -m_j).$$

In the basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta}, \hat{g}_j)$ the relation matrices are

$$D_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 6 & -1 & 1 & 0 & 2 \\ -6 & -1 & -2 & 0 & -4 \\ -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 \end{pmatrix}, \quad D_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & -1 & -1 & 0 & -3 \\ -6 & 1 & -1 & 0 & 3 \\ 3 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -4 \end{pmatrix}.$$

Integral row and column reduction gives $\text{diag}(1, 1, 1, 0, 0)$ for both, so $H_1(S_j; \mathbb{Z}) = \mathbb{Z}^2$ without torsion. Since S_j is a free finite quotient of a torus, $e(S_j) = 0$. The finite unramified covering $\pi_j : F \rightarrow S_j$ satisfies $\pi_j^* K_{S_j} = K_F \cong \mathcal{O}_F$; taking the norm gives $K_{S_j}^{m_j} \cong \mathcal{O}_{S_j}$.

Now $b_0 = b_4 = 1$ and $b_1 = b_3 = 2$ by Poincaré duality, while $0 = e(S_j) = 1 - 2 + b_2 - 2 + 1$ gives $b_2 = 2$. Also $H_3 \cong H^1 = \text{Hom}(H_1, \mathbb{Z})$ is free. Since $\text{Ext}(H_1, \mathbb{Z}) = 0$, the universal coefficient sequence gives $H^2 \cong \text{Hom}(H_2, \mathbb{Z})$; Poincaré duality identifies H^2 with H_2 , so H_2 is free as well. Hence

$$H_*(S_j; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z}^2, \mathbb{Z}), \quad (29)$$

and all these groups are torsion-free. Finally, K_{S_j} is torsion, so $w_2(S_j) \equiv c_1(K_{S_j}) \equiv 0 \pmod{2}$; Wu's formula $x^2 \equiv x \smile w_2(S_j) \pmod{2}$ shows that the unimodular intersection form on $H^2(S_j; \mathbb{Z})$ is even.

5.3.3 Specialization lattices at the elliptic points

Proposition 5.2 (Specialization lattices at the elliptic points). *For $0 \leq q \leq 3$, the map $\pi_j^* : H^q(S_j; \mathbb{Z}) \rightarrow (\Lambda^q V)^{T_j}$ is injective, with indices*

q	0	1	2	3
$j = 1$	1	3	1	1
$j = 2$	1	4	2	2

More explicitly,

$$\begin{aligned} \pi_1^* H^1(S_1) &= \langle 3\gamma, \eta_1 \rangle, & \pi_2^* H^1(S_2) &= \langle 4\gamma, \eta_2 \rangle, \\ \pi_1^* H^2(S_1) &= (\Lambda^2 V)^{T_1}, & \pi_2^* H^2(S_2) &= \{a\gamma\eta_2 + bq : a \equiv b \pmod{2}\}. \end{aligned}$$

For $q = 3$, the second cokernel is generated by $[\gamma uw]$.

Proof. Injectivity follows from the transfer identity $\pi_{j*} \pi_j^* = m_j \text{id}$ and the torsion-freeness of $H^q(S_j; \mathbb{Z})$. The extension

$$1 \rightarrow \Lambda \rightarrow \Gamma_j \rightarrow \mathbb{Z}/m_j \rightarrow 1$$

has class $[v_j] \in \Lambda^{A_j}/\Sigma_j \Lambda$, where $\Sigma_j = I + A_j + \cdots + A_j^{m_j-1}$. The edge map in the Lyndon–Hochschild–Serre spectral sequence gives

$$\pi_j^* H^1(S_j) = \{\xi \in V^{T_j} : m_j \mid \xi(v_j)\}.$$

Since $\eta_1(v_1) = \eta_2(v_2) = 0$ and $\gamma(v_1) = 1, \gamma(v_2) = -1$, the two degree-one formulas follow.

The degree-two invariants have basis $\langle \gamma\eta_j, q \rangle$, and

$$(\gamma\eta_1) \wedge q = 3\gamma \wedge u \wedge w \wedge \delta, \quad (\gamma\eta_2) \wedge q = 2\gamma \wedge u \wedge w \wedge \delta, \quad q^2 = 12\gamma \wedge u \wedge w \wedge \delta.$$

The intersection form on $H^2(S_j)$ is unimodular, while pullback along the covering multiplies intersection numbers by m_j . Comparing discriminants -9 and -4 gives indices 1 and 2. For $j = 2$, choose the section $s(r) = \widehat{g}_2^r$, $0 \leq r < 4$, of $\Gamma_2 \rightarrow \mathbb{Z}/4$. Its factor set is

$$\kappa(r, s) = \begin{cases} 0, & r + s < 4, \\ v_2, & r + s \geq 4, \end{cases}$$

because $\widehat{g}_2^4 = t_{v_2}$. Substitution in the cyclic bar resolution identifies the degree-two transgression on a decomposable invariant 2-form α with contraction $[\alpha] \mapsto [\iota_{v_2}\alpha]$, modulo the norm lattice $\langle 4\gamma, 2\eta_2 \rangle$. Hence

$$d_2(\gamma\eta_2) = -[\eta_2] \in \langle \gamma, \eta_2 \rangle / \langle 4\gamma, 2\eta_2 \rangle,$$

Thus $\gamma\eta_2$ is not in the image. The index-two sublattice must be $\langle 2\gamma\eta_2, a\gamma\eta_2 + q \rangle$ with $a = 0$ or 1 . The canonical class of S_2 is torsion and H^2 is torsion-free, hence $w_2(S_2) = 0$ and the intersection form is even. If $x = \gamma\eta_2$, then $x^2 = 0$, $xq = 2\gamma \wedge u \wedge w \wedge \delta$, and $q^2 = 12\gamma \wedge u \wedge w \wedge \delta$. Therefore the square of $ax + q$ upstairs is $(4a + 12)\gamma \wedge u \wedge w \wedge \delta$, and its square downstairs is $(4a + 12)/4 = a + 3$. Evenness excludes $a = 0$ and forces $a = 1$. The degree-three index is obtained from the perfect pairing $H^1 \times H^3 \rightarrow \mathbb{Z}$. Bases of degree-three invariants are

$$\begin{aligned} (\Lambda^3 V)^{T_1} &= \langle \gamma uw, \zeta_1 \rangle, & \zeta_1 &= -uw\delta + 2\gamma w\delta + 4\gamma u\delta, \\ (\Lambda^3 V)^{T_2} &= \langle \gamma uw, \zeta_2 \rangle, & \zeta_2 &= -uw\delta + 3\gamma w\delta + 3\gamma u\delta. \end{aligned}$$

The absolute determinants of the pairing matrices between the degree-one basis (γ, η_j) and these degree-three bases are 3 and 2, respectively. If the degree-one and degree-three image lattices have indices $k_j^{(1)}, k_j^{(3)}$, then the pairing upstairs is m_j times the pairing downstairs. Comparing determinants of the two rank-two pairings gives

$$k_j^{(1)} k_j^{(3)} |\det P_j| = m_j^2.$$

Substitution of $(k_1^{(1)}, k_2^{(1)}) = (3, 4)$ yields $k_1^{(3)} = 1, k_2^{(3)} = 2$. For $j = 2$, the class η_2 already belongs to the degree-one image, whereas $\langle \eta_2, \gamma uw \rangle = -2$ is not divisible by $m_2 = 4$. The pairing of two pullback classes must be divisible by 4, so γuw is not in the degree-three image; its class generates the order-two cokernel. $\square$

5.4 Nearby cycles at the cusp: the integral invariant-cycle theorem

This section proves, in the degrees needed later, that specialization at the cusp

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \longrightarrow H^q(F; \mathbb{Z})$$

gives precisely the T_0 -invariants, and that its image is saturated. This is essential for the integral Leray calculation below; equality of ranks over $\mathbb{Q}$ would not suffice.

Theorem 5.3 (Invariant cycles at the cusp). *Let $j : \Delta^* \hookrightarrow \Delta$. For $0 \leq q \leq 3$, in a neighborhood of 0 the natural map*

$$R^q f_{0*} \mathbb{Z} \longrightarrow j_* j^* R^q f_{0*} \mathbb{Z}$$

is an isomorphism. Equivalently, for these degrees,

$$\mathrm{sp}^q : H^q(W; \mathbb{Z}) \xrightarrow{\sim} H^q(F; \mathbb{Z})^{T_0}$$

is an isomorphism of integral lattices.

5.4.1 Local Milnor fibers

Let $\psi = \psi_{f_0} \mathbb{Z}_{N_0}$ be the nearby-cycle complex [4, 8]. Proper base change gives

$$\mathbb{H}^q(W; \psi) \cong H^q(F; \mathbb{Z}),$$

and the edge map $\mathbb{Z}_W \rightarrow \psi$ is precisely specialization. The three local models of Section 4.1 are

$$t_c = z_0, \quad t_c = z_0 z_1, \quad t_c = z_0 z_1 z_2.$$

Their Milnor fibers have the homotopy types of a point, S^1 , and T^2 , respectively. Hence

$$\begin{aligned} \mathcal{H}^0 \psi &= \mathbb{Z}_W, \\ \mathcal{H}^1 \psi &=: \mathcal{V}, & \mathcal{V} \text{ is supported on the double locus } D, \\ \mathcal{H}^2 \psi &= \mathbb{Z}_P \oplus \mathbb{Z}_Q, & \mathcal{H}^{\geq 3} \psi = 0. \end{aligned}$$

Write $\bar{C}_1, \bar{C}_2, \bar{C}_3 \cong \mathbb{P}^1$ for the three double curves, $\iota_k : \bar{C}_k \hookrightarrow W$ for their inclusions, and $\bar{C}_k^\circ = \bar{C}_k \setminus \{P, Q\}$. On $\bar{C}_k^\circ$, $\mathcal{V}$ is a rank-one constant sheaf. Label the local branches W_+, W_- and orient the vanishing circle by increasing $\arg z_+$. At a triple point with $t_c = z_0 z_1 z_2$,

$$H_1(\text{MF}) = \{(a_0, a_1, a_2) \in \mathbb{Z}^3 : a_0 + a_1 + a_2 = 0\}.$$

Restriction along the double curve $W_i \cap W_j$ is represented by $e_i - e_j$. To make the quotient sign check finite, list the six boundary directions cyclically as

$$r_0 = e_1, r_1 = e_2, r_2 = e_2 - e_1, r_3 = -e_1, r_4 = -e_2, r_5 = e_1 - e_2.$$

The side pairing is $r_k \leftrightarrow r_{k+3}$. Translate each of the six adjacent A_2 triangles back to the vertex 0 and preserve the branch order induced by the clockwise base meridian. The three upper triangles give the first list below and the three lower triangles give the second; this is the entire chart-by-chart check. For a stalk functional $\phi \in H^1(\text{MF}; \mathbb{Z}) = \text{Hom}(H_1(\text{MF}), \mathbb{Z})$, let n_k denote its restriction along the k th double curve. If the three branches at P are b_0, b_1, b_2 , the ordered branch pairs for the three curves are

$$(b_0, b_1), \quad (b_0, b_2), \quad (b_1, b_2);$$

and hence

$$(n_1, n_2, n_3) = (\phi(e_0 - e_1), \phi(e_0 - e_2), \phi(e_1 - e_2)).$$

If the branches at Q are c_0, c_1, c_2 , the opposite-edge gluing changes the ordered branch pairs to

$$(c_2, c_1), \quad (c_0, c_1), \quad (c_0, c_2),$$

and hence

$$(n_1, n_2, n_3) = (\phi(e_2 - e_1), \phi(e_0 - e_1), \phi(e_0 - e_2)).$$

At both points the unique relation is $n_1 - n_2 + n_3 = 0$, confirming that the signs at the two triple points agree. Therefore there is a short exact sequence

$$0 \rightarrow \mathcal{V} \rightarrow \bigoplus_{k=1}^3 \iota_{k*} \mathbb{Z}_{\bar{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \rightarrow 0, \quad \delta_1(n_1, n_2, n_3) = (s, s), \quad s = n_1 - n_2 + n_3. \quad (30)$$

Away from P, Q this is just the identity description of the rank-one vanishing-cycle sheaf on one double curve. At either triple point its stalk is the kernel of the displayed alternating-sum map, which is the rank-two group $\{(a_0, a_1, a_2) : a_0 + a_1 + a_2 = 0\}$. Thus exactness holds at every stalk.

5.4.2 Integral groups on the E_2 page

First compute $H^*(W; \mathbb{Z})$. The normalization $\nu : dP_6 \rightarrow W$ gives the resolution

$$0 \rightarrow \mathbb{Z}_W \rightarrow \nu_* \mathbb{Z}_{dP_6} \rightarrow \bigoplus_{k=1}^3 \iota_{k*} \mathbb{Z}_{\tilde{C}_k} \xrightarrow{\delta_1} \mathbb{Z}_P \oplus \mathbb{Z}_Q \rightarrow 0. \quad (31)$$

Here the first nontrivial arrow sends a locally constant function on the normalization to the differences of its restrictions to the two preimages of each double curve; the last arrow is the same alternating sum at P and Q . At a generic double point its kernel says that the two branch values agree, and at a triple point the three pairwise differences form the exact augmented Čech complex of three branches. Hence (31) is exact stalk by stalk. In cyclic order, the dP_6 hexagon is

$$E_1, H - E_1 - E_2, E_2, H - E_2 - E_3, E_3, H - E_1 - E_3.$$

In degree two, the difference map from $H^2(dP_6) = \mathbb{Z}^4$ to the three copies of $\mathbb{P}^1$ is

$$aH - \sum b_i E_i \mapsto (s, -s, s), \quad s = b_1 + b_2 + b_3 - a.$$

Its image is a saturated rank-one sublattice. Compute (31) row by row. In ordinary cohomological degree 0, the resolution gives the complex

$$\mathbb{Z} \longrightarrow \mathbb{Z}^3 \xrightarrow{\delta_1} \mathbb{Z}^2.$$

The first arrow is the difference of the restrictions to the two branches and is zero. The second has kernel $\mathbb{Z}^2$, image $\mathbb{Z}(1, 1)$, and cokernel $\mathbb{Z}$. In ordinary degree 2, there is only

$$\mathbb{Z}^4 \xrightarrow{L \mapsto (s, -s, s)} \mathbb{Z}^3,$$

whose kernel is $\mathbb{Z}^3$ and cokernel is $\mathbb{Z}^2$. In ordinary degree 4, there is only $H^4(dP_6) = \mathbb{Z}$. Combining by total degree gives

source	H^0	H^1	H^2	H^3	H^4
degree-0 row	$\mathbb{Z}$	$\mathbb{Z}^2$	$\mathbb{Z}$	0	0
degree-2 row	0	0	$\mathbb{Z}^3$	$\mathbb{Z}^2$	0
degree-4 row	0	0	0	0	$\mathbb{Z}$

Thus

$$H^*(W; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}), \quad (32)$$

and all groups are torsion-free.

For (30), the long exact sequence in degree 0 gives $\ker \delta_1 = \mathbb{Z}^2$ and $\operatorname{coker} \delta_1 = \mathbb{Z}$. The degree-two cohomology of the three copies of $\mathbb{P}^1$ contributes another $\mathbb{Z}^3$, while points have no higher cohomology. Hence

$$H^*(W; \mathcal{V}) = (\mathbb{Z}^2, \mathbb{Z}, \mathbb{Z}^3, 0).$$

Including $\mathcal{H}^2\psi$, the nearby-cycle hypercohomology spectral sequence

$$E_2^{a,b} = H^a(W; \mathcal{H}^b\psi) \implies H^{a+b}(F; \mathbb{Z})$$

has rank table

$b = 2$	2	0	0	0	0
$b = 1$	2	1	3	0	0
$b = 0$	1	2	4	2	1
	$a = 0$	$a = 1$	$a = 2$	$a = 3$	$a = 4$

Every entry is free abelian.

5.4.3 Differentials, degree-three specialization, and saturation

The only potentially nonzero differentials are

$$\begin{aligned} d_2 : E_2^{0,1} &\rightarrow E_2^{2,0}, & d_2 : E_2^{1,1} &\rightarrow E_2^{3,0}, & d_2 : E_2^{2,1} &\rightarrow E_2^{4,0}, \\ d_2 : E_2^{0,2} &\rightarrow E_2^{2,1}, & d_3 : E_3^{0,2} &\rightarrow E_3^{3,0}. \end{aligned}$$

Denote their ranks, in order, by $\alpha, \beta, \gamma, \delta, \epsilon$. Since $H^q(F) = \Lambda^q \mathbb{Z}^4$ has ranks 1, 4, 6, 4, 1, comparison in each total degree gives

$$\alpha = \gamma = 0, \quad \beta + \delta + \epsilon = 1.$$

Thus $E_2^{q,0} = H^q(W)$ survives except possibly in total degree 3.

For degree three, let W° denote the smooth open torus in W , namely the image of the dense torus in the normalization E_0 . Choose a compact T_f -orbit $K \cong T^2$ in W° , where $T_f = (\Lambda_{\text{tor}} \otimes \mathbb{R})/\Lambda_{\text{tor}}$. The map f_0 is a submersion near K . Choose a T_f -invariant metric and horizontally lift the two coordinate vector fields on the base. Compactness of K allows the neighborhood and radius to be reduced so that both flows exist for a uniform time. The vertical tangent space together with the two horizontal fields spans TN_0 . The inverse function theorem first gives a local diffeomorphism, and a finite shrinking using compactness makes it injective. Let $K' \subset W^\circ$ be the resulting T_f -invariant open neighborhood of K . One obtains a T_f -equivariant open embedding

$$\Theta : K' \times \Delta_r \hookrightarrow N_0, \quad f_0 \circ \Theta(w, t) = t, \quad \Theta(w, 0) = w.$$

Excision and the product chart give the naturality square

$$\begin{array}{ccc} H^3(W, W \setminus K) & \longrightarrow & H^3(W) \\ \downarrow \wr & & \downarrow \text{sp}^3 \\ H^3(F, F \setminus K_t) & \longrightarrow & H^3(F), \end{array}$$

where $K_t = \Theta(K, t)$. The bottom horizontal arrow is, by the Thom definition, the Gysin map of K_t in the general fiber. Thus the image of sp^3 contains the image of that Gysin map. The submanifold K_t is the linear subtorus $T_f \subset F$. Since Λ_{tor} is saturated, choose a complementary lattice Λ_B with $\Lambda = \Lambda_{\text{tor}} \oplus \Lambda_B$ and put $T_B = (\Lambda_B \otimes \mathbb{R})/\Lambda_B$. Then $F \cong T_f \times T_B$, and therefore

$$H^1(T_f) \xrightarrow{\iota_!} H^3(F), \quad a \longmapsto a \times [T_B]^\vee$$

is injective and has image a Künneth direct summand, hence a saturated rank-two sublattice. Therefore sp^3 has rank at least $2 = \text{rk } H^3(W)$; it is injective with saturated image. Consequently $\beta = \epsilon = 0$ and $\delta = 1$.

Finally, verify saturation degree by degree. For $q = 1$, the cokernel is $E_\infty^{0,1} \cong \mathbb{Z}^2$. For $q = 2$, the filtration quotients are $\mathbb{Z}$ and $\ker(d_2 : \mathbb{Z}^2 \rightarrow \mathbb{Z}^3)$, both free. Degree 3 was just proved, and in degree 0 specialization sends the constant 1 to the constant 1. Every extendable class is invariant under transport around the cusp, so

$$\text{Im } \text{sp}^q \subset (\Lambda^q V)^{T_0}.$$

For $0 \leq q \leq 3$, both sides are saturated sublattices of ranks 1, 2, 4, 2, so they are equal. This proves Theorem 5.3.

6 The integral Leray spectral sequence

Let $j : B^\circ \hookrightarrow B$, and let

$$\mathcal{L}^q = R^q f_* \mathbb{Z}|_{B^\circ}$$

be the integral local system with fiber $\Lambda^q V$ and monodromies T_1, T_2, T_0 . By Theorem 5.3, $R^q f_* \mathbb{Z} = j_* \mathcal{L}^q$ at p_0 . At p_1, p_2 , its stalks are the specialization lattices of Proposition 5.2. Therefore the quotient sheaf

$$Q^q := j_* \mathcal{L}^q / R^q f_* \mathbb{Z}$$

is a skyscraper sheaf, explicitly

$$\begin{array}{c|c} q & Q^q \\ \hline 1 & (\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2} \\ 2 & (\mathbb{Z}/2)_{p_2} \\ 3 & (\mathbb{Z}/2)_{p_2}. \end{array} \quad (33)$$

6.1 Sheaf cohomology on the base

For a finite-rank free local system $\mathcal{L}$, Poincaré duality gives

$$H^2(B, j_* \mathcal{L}) \cong H_c^2(B^\circ, \mathcal{L}) \cong H_0(B^\circ, \mathcal{L}) \cong M_G, \quad (34)$$

where M is a stalk and $G = \langle T_1, T_2 \rangle$. The first isomorphism follows because the quotient of $0 \rightarrow j! \mathcal{L} \rightarrow j_* \mathcal{L}$ is a skyscraper sheaf and has no degree-two cohomology. The second is Poincaré duality with local coefficients on the oriented real surface B° [3, Ch. V]. It is given by cap product with the locally finite fundamental class. The projection formula makes it natural for coefficient pairings; hence, if x is invariant, cup product corresponds to $x \smile [y] = [x \otimes y]$.

Using the short exact sequence $0 \rightarrow R^q f_* \mathbb{Z} \rightarrow j_* \mathcal{L}^q \rightarrow Q^q \rightarrow 0$, together with (33) and the integral-lattice computations of Section 2, we obtain

$$H^0(B, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\gamma, \quad H^0(B, R^2 f_* \mathbb{Z}) = \mathbb{Z} \cdot 2q, \quad (35)$$

$$H^0(B, R^3 f_* \mathbb{Z}) = \mathbb{Z} \cdot 2\gamma uw,$$

$$H^2(B, R^q f_* \mathbb{Z}) = (\Lambda^q V)_G = \mathbb{Z}[\bar{\delta}], \mathbb{Z}[\gamma\delta], \mathbb{Z}[uw\delta] \quad (q = 1, 2, 3). \quad (36)$$

The global invariant lattices in degrees one through three are $\mathbb{Z}\gamma, \mathbb{Z}q, \mathbb{Z}\gamma uw$. A degree-one class must be divisible by 3 at p_1 and by 4 at p_2 , so it lies in $12\mathbb{Z}\gamma$. In degrees two and three there is an order-two cokernel at p_2 , giving $2\mathbb{Z}q$ and $2\mathbb{Z}\gamma uw$. On the other hand, a skyscraper sheaf has no degree-two cohomology, so $H^2(B, R^q)$ agrees with $H^2(B, j_* \mathcal{L}^q)$. Equations (27)–(28) then give (36).

We also need $H^1(B, R^1) = H^1(B, R^2) = 0$. In the skyscraper long exact sequence, the maps on global sections are surjective: in degree one, $n\gamma \mapsto (n \bmod 3, n \bmod 4)$ is onto by the Chinese remainder theorem; in degree two, q maps to the nonzero element of $\mathbb{Z}/2$. Hence $H^1(B, R^q) \cong H^1(B, j_* \mathcal{L}^q)$ for $q = 1, 2$. The latter is parabolic cohomology. Its cocycles satisfy $c(t_i) \in (T_i - I)M$ for all three meridians. Subtracting a coboundary, arrange $c(t_2) = 0$; the residual gauge freedom is M^{T_2} . The cocycle formula for $t_0 = (t_1 t_2)^{-1}$ gives $c(t_0) = -T_0 c(t_1)$. Therefore

$$H^1(B, j_* \mathcal{L}^q) \cong \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}.$$

For $q = 1$,

$$(T_1 - I)V = \langle 2\gamma + u + w, 2\gamma - u \rangle, \quad (T_0 - I)V = \langle \gamma, u \rangle,$$

so the intersection is $\mathbb{Z}(2\gamma - u) = \mathbb{Z}(T_1 - I)\eta_2$. For $q = 2$,

$$(\Lambda^2 T_1 - I)\Lambda^2 V = \langle \gamma u, \gamma w, 2u\delta + w\delta, uw + u\delta - w\delta - 6\gamma\delta \rangle,$$

$$(\Lambda^2 T_0 - I)\Lambda^2 V = \langle \gamma u, \gamma w + u\delta \rangle.$$

Comparison of the coefficients of $uw, u\delta, w\delta$ shows that the intersection can only be $\mathbb{Z}\gamma u = \mathbb{Z}(\Lambda^2 T_1 - I)(\gamma\eta_2)$; Thus both quotients vanish.

6.2 The three d_2 differentials

Since the base has complex dimension one, the Leray spectral sequence

$$E_2^{a,b} = H^a(B, R^b f_* \mathbb{Z}) \implies H^{a+b}(X; \mathbb{Z})$$

has only $a = 0, 1, 2$. The only differentials needed in total degree at most three are $d_2 : E_2^{0,b} \rightarrow E_2^{2,b-1}$ for $b = 1, 2, 3$; possible higher-row differentials do not enter the argument because the upper homology will follow from Poincaré duality.

Set

$$p := 12\ell_0 - 4\ell_1 - 3\ell_2 = -1$$

for the fixed boundary data (25).

Proposition 6.1. *If ω generates $H^2(B; \mathbb{Z})$, then, with one common overall sign,*

$$d_2(12\gamma) = p\omega, \tag{37}$$

$$d_2(2q) = p\omega\bar{\delta}, \tag{38}$$

$$d_2(2\gamma uw) = p\omega[\gamma\delta]. \tag{39}$$

Proof. The differential $d_2(12\gamma)$ is the obstruction to gluing the local degree-one classes to a global class on X . The class lifts over N_0 because $\gamma \in V^{T_0}$ and the cusp meridian vanishes in N_0 . It lifts over N_j because $12\gamma \in \pi_j^* H^1(S_j)$. Any lift Θ_j takes the values

$$\Theta_1(\hat{\rho}_1) = 4\ell_1, \quad \Theta_2(\hat{\rho}_2) = 3\ell_2.$$

Indeed, Θ_j restricts to 12γ on the fiber subgroup Λ , and the relation $\hat{\rho}_j^{m_j} = t_{v_j}$ forces

$$m_j \Theta_j(\hat{\rho}_j) = 12\gamma(v_j) = 12\ell_j;$$

Substitution of $m_1 = 3, m_2 = 4$ gives the displayed values. The untranslated cusp gluing contributes $12\ell_0 = 0$, so the sum of the differences on the three annular overlaps is

$$12\ell_0 - 4\ell_1 - 3\ell_2 = p.$$

More formally, cover B by the union U of three small disks and the complementary pair-of-pants neighborhood V . Modulo the sum-zero sublattice $H^1(V)$, the group $H^1(U \cap V) = \mathbb{Z}^3$ is identified by summing the three meridian values with $H^2(B) = \mathbb{Z}\omega$. This proves (37).

Write

$$d_2(2q) = p'\omega\bar{\delta}, \quad d_2(2\gamma uw) = p''\omega[\gamma\delta].$$

Apply multiplicativity of the spectral sequence, naturality of (34), and the Leibniz rule to $(12\gamma)(2q) = 12(2\gamma uw)$, using $[q] = 12[\gamma\delta]$. This gives

$$p'' = 2p - p'.$$

Next use $(2q)(2\gamma uw) = 0$. With the declared fiber orientation $\gamma \wedge u \wedge w \wedge \delta = \gamma \wedge u \wedge w \wedge \delta$, the two coefficient pairings have signs

$$\bar{\delta} \wedge 2\gamma uw = -2\gamma \wedge u \wedge w \wedge \delta, \quad 2q \wedge [\gamma\delta] = 2\gamma \wedge u \wedge w \wedge \delta.$$

The graded Leibniz rule therefore gives $-2p' + 2p'' = 0$, so $p' = p''$. Combining this with $p'' = 2p - p'$ yields $p' = p'' = p$. This fixes the relative signs and proves all three formulas. $\square$

Since $p = -1$, all three displayed d_2 maps are isomorphisms. To make the passage from E_∞ explicit, the filtration pieces in total degrees one, two, and three are

$$\begin{aligned} H^1(X) : \quad & E_\infty^{1,0} = 0, & E_\infty^{0,1} = \ker d_2^{0,1} = 0; \\ H^2(X) : \quad & E_\infty^{2,0} = \operatorname{coker} d_2^{0,1} = 0, & E_\infty^{1,1} = 0, & E_\infty^{0,2} = \ker d_2^{0,2} = 0; \\ H^3(X) : \quad & E_\infty^{2,1} = \operatorname{coker} d_2^{0,2} = 0, & E_\infty^{1,2} = 0, & E_\infty^{0,3} = \ker d_2^{0,3} = 0. \end{aligned}$$

Here $E_\infty^{1,1} = E_\infty^{1,2} = 0$ uses the two parabolic-cohomology calculations above. Since every associated graded piece is zero, $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0$. We now convert the cohomology statement to homology term by term, so that no torsion is hidden. Theorem 5.1 gives $\pi_1(X) = 1$, hence $H_1(X) = \pi_1(X)_{\text{ab}} = 0$. The universal coefficient exact sequence in degree two is

$$0 = \operatorname{Ext}(H_1(X), \mathbb{Z}) \longrightarrow H^2(X; \mathbb{Z}) \longrightarrow \operatorname{Hom}(H_2(X), \mathbb{Z}) \longrightarrow 0,$$

Thus $H_2(X)$ can have only finite torsion. In degree three, the universal coefficient sequence injects $\operatorname{Ext}(H_2(X), \mathbb{Z})$ into $H^3(X; \mathbb{Z}) = 0$. For a finitely generated torsion group A , $\operatorname{Ext}(A, \mathbb{Z}) \cong A$, so $H_2(X) = 0$. Finally, X is a compact complex threefold and therefore an oriented closed real six-manifold. Integral Poincaré duality gives

$$H_3(X) \cong H^3(X) = 0, \quad H_4(X) \cong H^2(X) = 0, \quad H_5(X) \cong H^1(X) = 0.$$

Therefore

Theorem 6.2.

$$H_k(X; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \leq k \leq 5. \end{cases}$$

Thus X is an integral homology six-sphere.

6.3 Recognition as the standard S^6 and transport of the complex structure

By Theorems 5.1 and 6.2, X is simply connected and has the integral homology of S^6 . Apply Hurewicz inductively: first $\pi_2(X) \cong H_2(X) = 0$; if $\pi_i(X) = 0$ for $i < k$, then Hurewicz gives $\pi_k(X) \cong H_k(X) = 0$ through $k = 5$, and finally $\pi_6(X) \cong H_6(X) = \mathbb{Z}$. A map $S^6 \rightarrow X$ representing a generator is therefore an isomorphism on all integral homology groups. Both spaces are simply connected, and smooth manifolds have the homotopy type of CW complexes, so the homological Whitehead theorem [6, Ch. 4] makes it a homotopy equivalence. Thus X is a smooth homotopy six-sphere.

In the sense of Kervaire–Milnor, X defines an element of the group Θ_6 of smooth homotopy spheres. Smale’s h -cobordism theorem [9] says that a homotopy sphere representing the zero element is diffeomorphic to the standard sphere. Kervaire–Milnor computed [7]

$$\Theta_6 = 0,$$

Thus the element represented by X is the standard sphere, and there is a diffeomorphism

$$\Phi : S^6 \xrightarrow{\sim} X.$$

If (U_α, z_α) is the holomorphic atlas of X , then

$$(\Phi^{-1}(U_\alpha), z_\alpha \circ \Phi)$$

is a complex-coordinate atlas on S^6 . Its transition functions are exactly those on X , and are therefore holomorphic. This proves Theorem 1.1.

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